Capstone Project
Machine Learning Engineer Nanodegree

Chunyan He
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Definition

Project Overview
Customer relationship management and personalised marketing are central problems in modern retail analytics. Companies increasingly rely on data-driven decision systems to deliver targeted promotions that maximise customer engagement while minimising marketing cost.
Company like Starbucks use loyalty programs to understand purchasing behaviour and influence customers through personalised offers. Digital promotions allow businesses to experiment with different incentive structures such as buy-one-get-one offers, discounts, and informational campaigns. Previous research shows that customers respond differently to promotions depending on demographic attributes, historical purchase behaviour, and communication channels. Personalised promotions can significantly increase conversion rate compared to uniform campaigns.
This project investigates how customer segmentation and predictive modelling can improve promotional targeting strategies.

Problem Statement
To identify which groups of customers are most responsive to different types of promotional offers and predict the likelihood that a customer will complete an offer.

Metrics
For classification models, ROC-AUC score will be used to measure model discrimination ability. Moreover, accuracy, precision, recall and F1 score will also be used to evaluate the model performance comprehensively.
Accuracy provides a general measure of overall correctness. Precision measures how effectively the model targets customers who are likely to complete an offer. Recall evaluates the model's ability to identify all potential responders. F1 score balances precision and recall, ensuring that the model achieves both efficient targeting and broad customer coverage.
For regression models, metrics such as R-squared (R^2) and Root Mean Squared Error (RMSE) will be used to evaluate model performance. These metrics help assess how accurately the model predicts the target variable and how well it explains the variation in the data.
R-squared measures the proportion of variance in the target variable that is explained by the model. A higher R-squared value indicates that the model is better at capturing the relationship between customer characteristics and the predicted outcome. RMSE evaluates the average magnitude of prediction errors by measuring the square root of the mean squared differences between actual and predicted values. Lower RMSE values indicate that the model's predictions are closer to the true values.

Analysis

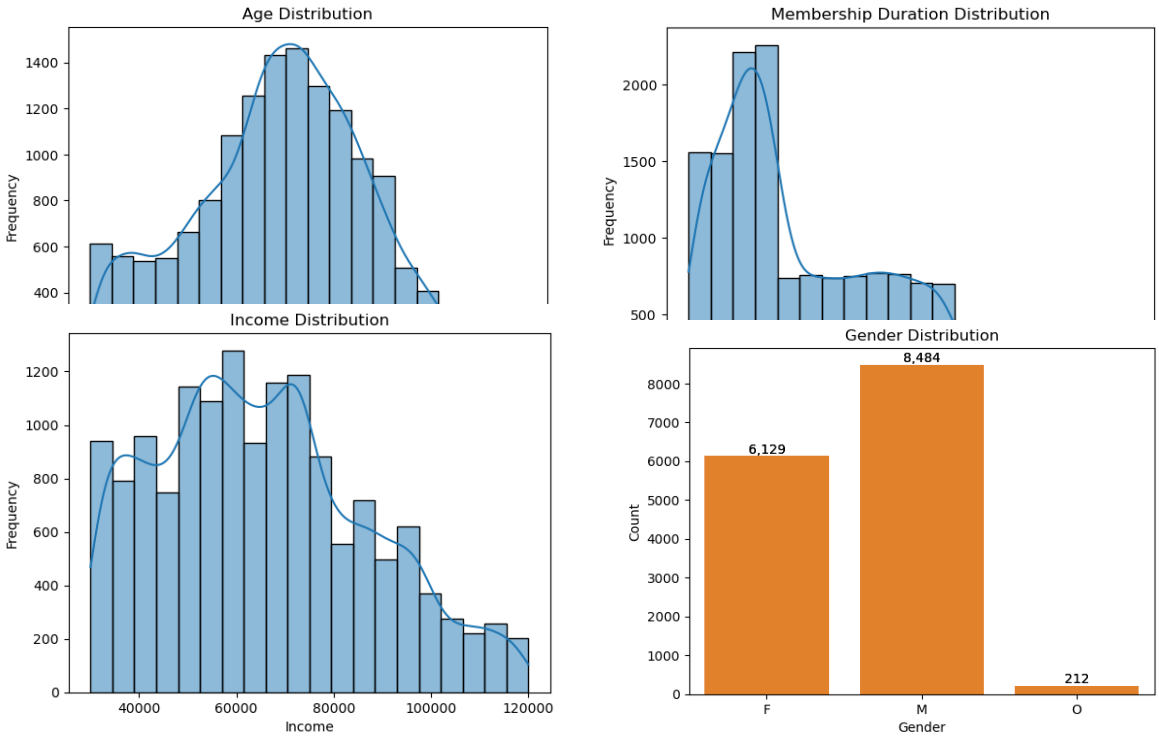
Data Exploration
Three datasets are provided:
1. profile.json: rewards program users (17000 users * 5 fields)
• gender: (categorical) M, F, O, or null
• age: (numeric) missing value encoded as 118
• id: (string/hash)
• became_member_on: (date) format YYYYMMDD
• income: (numeric)
2. portfolio.json: offers sent during 30-day test period (10 offers * 6 fields)
• reward: (numeric) money awarded for the amount spent
• channels: (list) web, email, mobile, social
• difficulty: (numeric) time for offer to be open, in days
• offer_type: (string) logo, discount, informational
• id: (string/hash)
3. transcript.json: event log (306648 events * 4 fields)
• person: (string/hash)

• event: (string) offer received, offer viewed, transaction, offer completed
• value: (dictionary) different values depending on event type
• offer id: (string/hash) not associated with any 'transaction'
• amount: (numeric) money spent in 'transaction'
• reward: (numeric) money gained from 'offer completed'
• time: (numeric) hours after start of test

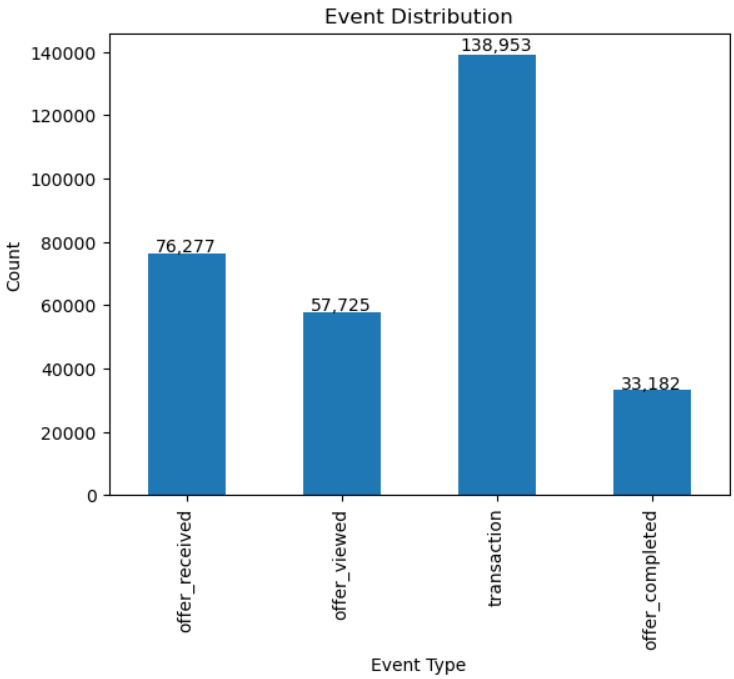
Exploratory Visualisation

The bar plot below is created with the portfolio data, which contains 10 different offers. Among the 10 offers, four are buy-one-get-one, two is informational offer and rest are the discount offer. And the offers are sent through 4 different channels: web, social, mobile and email.

Age distribution, membership duration distribution, income distribution and gender distribution are generated from the profile data. As showing in the plots, majority of the members aged from 40 to 70, most of the members have been holding their memberships for 7 to 8 years, income distributed around the range of 5000 to 8000, and male customers are 15.9% more than the female customers.



The event distribution is created from the transcript data. Which shows that 75.7% offers were viewed and 43.5% offers were completed.



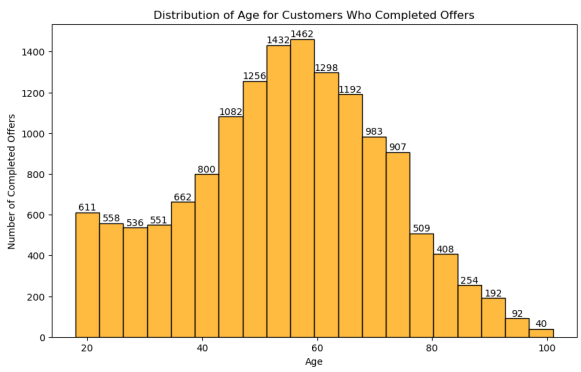
Correlation analysis was implemented on the best_channel DataFrame, which indicates that email is the most effective channel for promoting offers, followed by web and mobile.

	offer_completed	web	mobile	social	email
offer_completed	1.000000	0.329788	0.290016	0.219719	0.333833
web	0.329788	1.000000	0.704478	0.402283	0.809292
mobile	0.290016	0.704478	1.000000	0.748289	0.913779
social	0.219719	0.402283	0.748289	1.000000	0.683771
email	0.333833	0.809292	0.913779	0.683771	1.000000

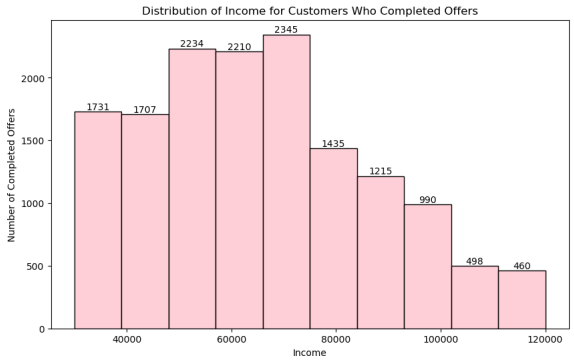
Correlation analysis were implemented on the best_offer DataFrame, which shows that discount offer is the most popular one which customers are more likely to complete. Followed by buy-one-get one.

	offer_completed	bogo	discount	informational
offer_completed	1.000000	0.204630	0.262267	-0.110041
bogo	0.204630	1.000000	-0.300144	-0.166384
discount	0.262267	-0.300144	1.000000	-0.163686
informational	-0.110041	-0.166384	-0.163686	1.000000

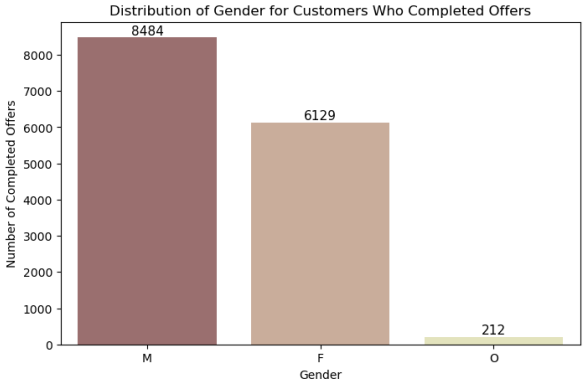
Customers' age, income and gender were analysed on the target_customer DataFrame in this project to determine the target audience. According to the analysis, customers who aged between 45 - 65 years old are more likely to complete the offers.



Most customers have decent earning. The top tier customers earn more than 100k per year, and the majority of the customers who usually complete offers earns around 50,000 - 70,000 per year.



Gender wise, male customer complete more offers than female.



Algorithms and techniques

Correlation:

- It is a statistical measure that describes the strength and direction of the relationship between two variables.
- It ranges from -1 to 1. While 1 represents positive relationship, -1 represents negative relationship.
- It will be used to identify the most effective channel and most popular offer.

Linear Regression:

- It is a supervised learning algorithm that models the relationship between one or more independent variables and a continuous dependent variable. It assumes a linear relationship and estimates the best-fitting line through the data by minimising the sum of square residuals.

Random Forest Regressor:

- it is an ensemble learning method that constructs a large number of decision trees at training time and outputs the mean prediction or majority vote of the individual trees. It combines two core techniques: Bootstrap Aggregating and Random Feature Subsampling.
- It provides built-in feature importance scores based on the total reduction in impurity contributed by each feature across all trees.

Gradient Boosting Regressor:

- It is an ensemble technique that builds trees sequentially, where each new tree is trained to correct the residual errors of the previous ensemble. unlike Random Forest, which builds trees in parallel and independently, Gradient Boosting is an iterative, additive process that directly optimise a differentiable loss function.

Logistic Regression:

- It is a classification algorithm, it models the probability that an instance belongs to a given class by applying the logistic function to a linear combination of features, mapping any real-valued input to a probability between 0 and 1.

Random Forest Classifier:

- It applies the same ensemble methodology as the Random Forest Regressor, but outputs a class label instead of a continuous value. Each tree casts a vote for a class, and the forest outputs the majority vote as the final prediction.

Gradient Boosting Regressor:

- It extends the gradient boosting framework to classification tasks by optimising classification-specific loss functions such as log-loss for binary tasks or multi-class log-loss for multi-class problems.

Benchmark

A majority class baseline classifier is established as the minimum performance threshold. Since the dataset contains customers who completed offers at an estimated rate of 57% based on the event distribution , a naive classifier that always predicts the majority class achieves the following:

Metric	Majority Class Baseline Value
Accuracy	57% - simply always predicting the majority class
Precision	0.571 - same as accuracy for a constant classifier
Recall	1 - always predicts positive, so captures all positives
F1 Score	0.727 - inflated by perfect recall buy low precision
ROC - AUC	0.5 - equivalent to random discrimination ability

All trained model must meaningfully exceed this baseline, particularly on ROC-AUC and F1 Score to justify their use over a trivial prediction strategy.
For the regression models, a mean predictor baseline yields an R squared of 0.00 by definition. All regression models must exceed R squared = 0.00 to demonstrate any predictive utility.

Methodology

Data Preprocessing

The data preprocessing and feature engineering are processed in the ‘preprocessing’ notebook.

- Unwrapped the value of ‘channels’ column in portfolio data which are originally lists
- Unwrapped the ‘value’ column in the transcript data
- Encoded categorical column ‘event’
- Dropped rows with ‘118’ as the value for the ‘age’ column
- The 3 datasets are merged by 'id' column in profile_data and 'person' column in transcript_data, then merged to portfolio data by 'id'(offer id) column.
- Then following datasets were created for checking the popularity of the offers, the effectiveness of different channels, and the target customers. Which are best_offer, best_channel, and target_customer respectively. target_customer DataFrame will be used to predict which customers should be targeted for the offers.

	offer_completed	web	mobile	social	email
0	0	0.0	0.0	0.0	0.0
1	0	0.0	0.0	0.0	0.0
2	0	1.0	1.0	0.0	1.0
3	0	1.0	1.0	0.0	1.0
4	0	0.0	0.0	0.0	0.0
...	...	...	...	...	...
272383	1	1.0	1.0	1.0	1.0
272384	0	0.0	0.0	0.0	0.0
272385	0	1.0	1.0	1.0	1.0
272386	0	1.0	1.0	0.0	1.0
272387	0	0.0	0.0	0.0	0.0

272388 rows × 5 columns

	offer_completed	bogo	discount	informational
0	0	0	0	0
1	0	0	0	0
2	0	1	0	0
3	0	0	0	1
4	0	0	0	0
...	...	...	...	...
272383	1	1	0	0
272384	0	0	0	0
272385	0	1	0	0
272386	0	0	0	1
272387	0	0	0	0

272388 rows × 4 columns

id	age	gender	income	member_days	offer_completed	offer_received
0009655768c64bdeb2e877511632db8f	33	M	72000.0	3243	3	5
0011e0d4e6b944f998e987f904e8c1e5	40	O	57000.0	2980	3	5
0020c2b971eb4e9188eac86d93036a77	59	F	90000.0	3656	3	5
0020ccbbb6d84e358d3414a3ff76cffd	24	F	60000.0	3404	3	4
003d66b6608740288d6cc97a6903f4f0	26	F	73000.0	3182	3	5
...	...	...	...	...	...	...
fff3ba4757bd42088c044ca26d73817a	69	F	83000.0	3822	3	6
fff7576017104bcc8677a8d63322b5e1	71	M	73000.0	3050	3	5
fff8957ea8b240a6b5e634b6ee8eafcf	71	M	56000.0	2940	0	3
fffad4f4828548d1b5583907f2e9906b	34	M	34000.0	3331	3	4
ffff82501cea40309d5fdd7edcca4a07	45	F	62000.0	3390	6	6

14825 rows × 6 columns

- After dropping the null values in the merged DataFrame, there are 14825 customers. 15,100 BOGO offer were completed, 16,970 discount offers were completed.

Implementation

All 6 models were implemented using Python’s scikit-learn library. The same train/test split was applied consistently across all models to ensure fair comparison. Default hyper parameters were used in the initial implementation, with refinement applied subsequently.

Component	Details
Language & Libraries	Python, scikit-learn, pandas, bumpy, matplotlib, seabird
Classification Models	LogisticRegression(max_iter=500)RandomForestClassifier(*n_estimators*=200, *random_state*=42) GradientBoostingClassifier(*n_estimators*=200, *learning_rate*=0.1, *random_state*=42)
Regression Models	LinearRegression() RandomForestRegressor(*n_estimators*=200, *random_state*=42) GradientBoostingRegressor(*n_estimators*=200, *random_state*=42)
Evaluation	Metrics computed on the held-put test set (20%). Feature importance extracted from ensemble models via feature importance attribute.
Complications	The high correlation between channel features introduced multicollinearity warnings in logistic Regression. This was mitigated by L2 regularisation. The class distribution was moderately imbalanced, contributing to high Recall but moderate Precision across all classifiers.

Refinement

Logistic Regression was tuned across four regularisation strengths (C = 0.01, 0.1, 1.0, 10.0). The results were remarkably stable, ROC-AUC held at 0.7782 across C = 0.1, 1.0, and 10.0, with only a negligible drop at C = 0.01 (0.7776). This flatness indicates that the model is not sensitive to regularisation strength on this dataset, and that the decision boundary is inherently well-behaved. C = 1.0 was retained as the final configuration as it is the standard default and showed no disadvantage.

Random Forest was tuned across n_estimators values of 50, 100, 150, and 200. A clear upward trend in ROC-AUC was observed as the forest grew from 0.7522 at 50 trees to 0.7597 at 200 trees. Though the gains diminished significantly after 100 trees. Accuracy and F1 Score followed the same pattern, plateauing around 150 to 200 trees. This confirms that 200 estimators is a reasonable stopping point where additional trees yield no meaningful improvement.

Gradient Boosting showed the most informative tuning behaviour. Across all combinations of learning rate (0.05, 0.1, 0.2) and max_depth (2, 3, 5), a consistent pattern emerged: shallower trees (depth=2) outperformed deeper ones at every learning rate, and slower learning rates (0.05) generally achieved higher ROC-AUC. The best configuration overall was lr=0.05, depth=2 (ROC-AUC = 0.8075), marginally outperforming the originally chosen lr=0.1, depth=3 (ROC-AUC = 0.8026). Deeper trees (depth=5) consistently degraded performance — particularly at higher learning rates — confirming that overfitting occurs when individual trees are too complex. Based on these results, the final Gradient Boosting model is updated to lr=0.05, depth=2.

```
=== Logistic Regression — C (Regularisation) Tuning ===
  C= 0.01 → Accuracy: 0.8132 | F1: 0.8957 | ROC-AUC: 0.7776
  C= 0.10 → Accuracy: 0.8121 | F1: 0.8946 | ROC-AUC: 0.7782
  C= 1.00 → Accuracy: 0.8128 | F1: 0.8949 | ROC-AUC: 0.7782
  C=10.00 → Accuracy: 0.8125 | F1: 0.8947 | ROC-AUC: 0.7782

=== Random Forest — n_estimators Tuning ===
  n_estimators= 50 → Accuracy: 0.7879 | F1: 0.8743 | ROC-AUC: 0.7522
  n_estimators=100 → Accuracy: 0.7943 | F1: 0.8784 | ROC-AUC: 0.7546
  n_estimators=150 → Accuracy: 0.7939 | F1: 0.8785 | ROC-AUC: 0.7583
  n_estimators=200 → Accuracy: 0.7939 | F1: 0.8787 | ROC-AUC: 0.7597

=== Gradient Boosting — learning_rate & max_depth Tuning ===
  lr=0.05 | depth=2 → Accuracy: 0.8185 | F1: 0.8969 | ROC-AUC: 0.8075
  lr=0.05 | depth=3 → Accuracy: 0.8162 | F1: 0.8957 | ROC-AUC: 0.8068
  lr=0.05 | depth=5 → Accuracy: 0.8101 | F1: 0.8910 | ROC-AUC: 0.8007
  lr=0.1 | depth=2 → Accuracy: 0.8162 | F1: 0.8955 | ROC-AUC: 0.8063
  lr=0.1 | depth=3 → Accuracy: 0.8121 | F1: 0.8927 | ROC-AUC: 0.8026
  lr=0.1 | depth=5 → Accuracy: 0.8054 | F1: 0.8872 | ROC-AUC: 0.7886
  lr=0.2 | depth=2 → Accuracy: 0.8162 | F1: 0.8953 | ROC-AUC: 0.8022
  lr=0.2 | depth=3 → Accuracy: 0.8081 | F1: 0.8896 | ROC-AUC: 0.7938
  lr=0.2 | depth=5 → Accuracy: 0.7960 | F1: 0.8804 | ROC-AUC: 0.7731
```

Results

Model Evaluation and Validation

All models were evaluated on the same held-out 20% test set using a fixed random seed (random_state=42), ensuring that performance differences between models reflect genuine differences in predictive ability rather than data sampling variation. Stratified splitting was applied for the classification task to preserve class proportions across the train and test sets.

For classification, Logistic Regression achieved an accuracy of 0.8128, F1 Score of 0.8949, and ROC-AUC of 0.7782. Despite being the simplest model, it delivered highly competitive performance, particularly on Recall (0.9854), capturing nearly every actual offer completer in the test set. Random Forest was the weakest of the three classifiers, achieving a ROC-AUC of 0.7597 and accuracy of 0.7939. Its Precision was the highest of the three (0.8383), but it missed more true positives than the other models. Gradient Boosting, configured with lr=0.05 and depth=2 after refinement, achieved the strongest overall result with a ROC-AUC of 0.8075, which the best discrimination across all thresholds, alongside an accuracy of 0.8185 and F1 Score of 0.8969.

For regression, all three models struggled to explain variance in the offer_completed_rate target. Linear Regression and Random Forest Regressor performed almost identically (R^2 = 0.1641 and 0.1634 respectively), which is a telling result — the additional complexity of Random Forest provided no benefit over a simple linear model. Gradient Boosting Regressor was the clear winner at R^2 = 0.2531 and RMSE = 0.2875, but even this result means roughly 75% of the variance in offer completion rate remains unexplained. This points to a data limitation rather than a modelling failure: demographic features alone (age, income, gender, member_days) do not contain sufficient signal to reliably predict how often a customer completes an offer.

```
--- Logistic Regression ---
Accuracy: 0.8128161888701517
Precision: 0.8195697432338653
Recall: 0.985398414685023
F1 Score: 0.8948664519795416
ROC-AUC: 0.7782314454100488


--- Random Forest ---
Accuracy: 0.793929173693086
Precision: 0.8382575757575758
Recall: 0.9232373800584064
F1 Score: 0.8786976374826285
ROC-AUC: 0.7597337781381657


--- Gradient Boosting ---
Accuracy: 0.812141652613828
Precision: 0.829041487839771
Recall: 0.9670421360033375
F1 Score: 0.8927402272289621
ROC-AUC: 0.8025844365315798
```

```
--- Linear Regression ---
R2 Score: 0.1640868892140256
RMSE: 0.30418156215667314


--- Random Forest Regressor ---
R2 Score: 0.16338944683231005
RMSE: 0.3043084323285575


--- Gradient Boosting Regressor ---
R2 Score: 0.25310278330514313
RMSE: 0.28752971772487057
```

Justification

Classification: All three classifiers comfortably exceeded the majority class baseline (ROC-AUC = 0.500, F1 = 0.727). The best model Gradient Boosting with lr=0.05, depth=2 achieved a ROC-AUC of 0.8075, representing a 61.5% relative improvement over the baseline. The F1 Score of 0.8969 exceeds the baseline by 16.9 percentage points. These gains are practically significant: a model that correctly identifies offer completers with 80%+ discriminative ability enables Starbucks to meaningfully personalise promotions, reduce wasted spend on non-converters, and improve overall campaign return.

A particularly noteworthy finding is how well Logistic Regression performed relative to the ensemble methods. Its ROC-AUC (0.7782) and F1 (0.8949) came close to Gradient Boosting despite being orders of magnitude simpler. In a production context where model interpretability and inference speed matte, such as real-time offer targeting in a mobile app, Logistic Regression is a compelling alternative that sacrifices very little performance for substantial gains in transparency.

Regression: While all regression models beat the mean predictor baseline (R^2 = 0.00), the best achievable R^2 of 0.2531 is insufficient for reliable prediction of offer completion rate. The near-identical performance of Linear Regression and Random Forest Regressor further suggests that the relationship between demographic features and completion rate is approximately linear and weak. The most productive next step is not further model tuning, but rather feature enrichment: incorporating offer channel mix, offer type history, transaction frequency, and time-since-last-purchase would likely yield substantially higher R^2 and unlock the regression task's practical utility.